Data Analytics Capstone

Data-Driven Dashboard Development for Carmela Industries

FINAL REPORT

Submitted By:

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1. Executive Summary

This project focused on designing a scalable, automated employee engagement dashboard for **Carmela Industries** to support data-driven decision-making. The primary objective was to transform survey-based employee feedback into actionable insights across four key engagement dimensions: **Happiness, Personal Motivation, Company Motivation**, and **Workplace Relationships.**

Due to constraints in accessing real employee data, specifically, the variability of survey questions and restricted access to BambooHR, this project adopted a **prototype-driven approach using simulated data**. The simulated dataset was carefully designed to mirror Carmela Industries' actual survey structure and themes, ensuring relevance and applicability.

The resulting Power BI dashboard demonstrates how Carmela can:

- Monitor employee engagement in real time
- Identify department-level performance gaps
- Track key indicators such as Net Promoter Score (NPS)
- Enable leadership to make faster, data-informed decisions

The project concludes that **full automation is feasible**, provided that standardized survey structures and system integrations (e.g., BambooHR API) are implemented.

2. Introduction & Business Context

Carmela Industries is focused on improving employee engagement through regular internal surveys. These surveys capture employee perceptions across multiple dimensions, providing valuable insights into workplace satisfaction and organizational effectiveness.

However, the organization currently faces key challenges:

- Survey questions change across cycles
- Data is not standardized for analysis
- Reporting is manual and time-consuming
- No centralized dashboard exists for real-time insights

These challenges limit Carmela's ability to:

- Track engagement trends over time
- Compare departments effectively
- Make timely, evidence-based decisions

This project builds on prior work to design a scalable, automated dashboard solution aligned with Carmela's engagement strategy.

The variability in survey questions across cycles initially posed a challenge for automation. However, this project demonstrates that by restructuring the data model into a flexible, theme-based framework, Carmela Industries can achieve automation without requiring identical survey questions.

3. Methodology

3.1 Approach

A design science and prototyping approach was used:

- Recreate Carmela's survey structure
- Simulate realistic response data
- Build a data model in Power BI
- Develop dashboards reflecting real business use cases

3.2 Tools Used

- **Microsoft Power BI** – Dashboard development
- **Microsoft Excel** – Data storage and preprocessing
- **Google Forms** – Data collection simulation
- Power Query – Data transformation

3.3 Data Limitations

- Lack of Real Employee Data
- No direct access to BambooHR survey data
- Simulated dataset used instead

3.4 Dynamic Survey Structure

- Questions vary across survey cycles
- Makes automation difficult

3.5 Small Sample Size

- Responses collected from a controlled group
- Not representative of the actual workforce

3.6 Justification of Approach

Despite these limitations, the methodology is valid because:

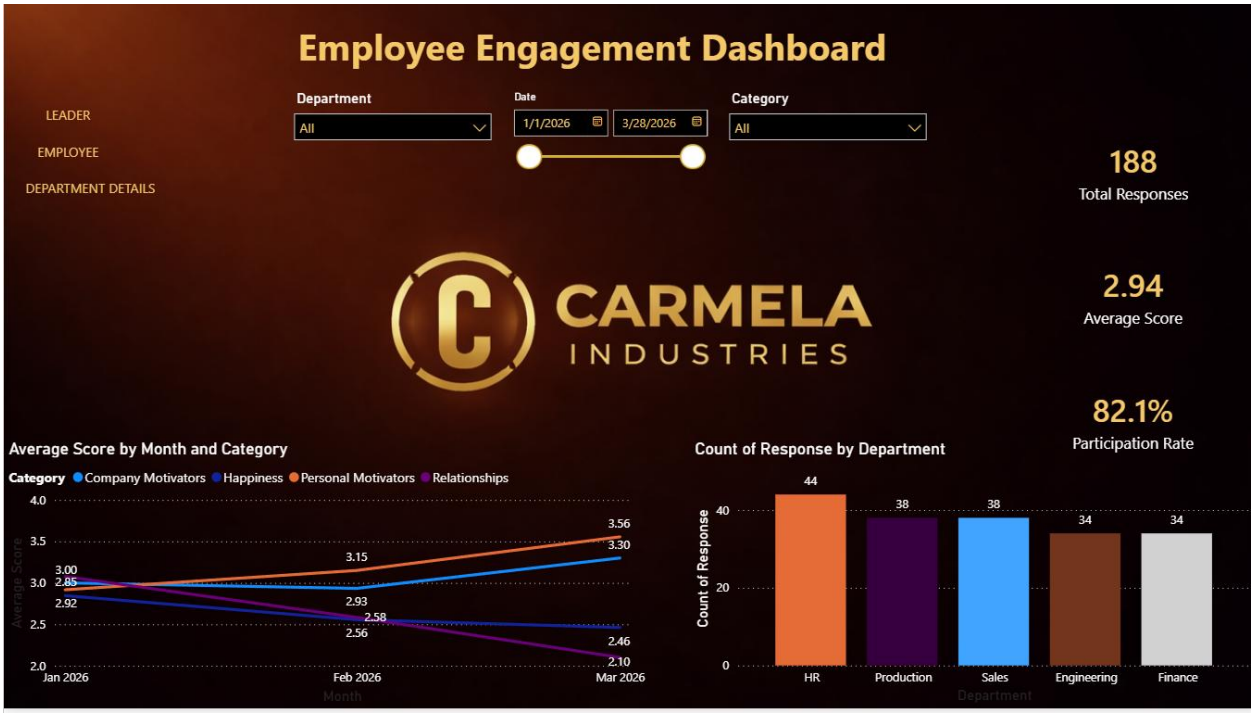
- The survey structure mirrors real organizational design

- The dashboard tests functionality, not outcomes
- The solution is scalable and ready for real data integration

4. Findings & Analysis

4.1 Engagement Overview

The analysis identified four key engagement drivers:



4.2 Key Insights

Key insights from the Employee Engagement Dashboard for Carmela Industries show the overall engagement levels, participation trends, departmental differences, and changes across key engagement dimensions: **Happiness, Personal Motivation, Company Motivators**, and **Workplace Relationships.**

The organization recorded an average engagement score of **2.94** out of **5,** indicating a moderate level of employee engagement. While employees are not disengaged, the score suggests that engagement is not strong enough to drive optimal performance and satisfaction.

The dashboard shows a **participation rate of 82.1%**, which is considered high for employee surveys.

A significant decline is observed in:

- **Workplace Relationships** (sharp drop across months)
- **Happiness levels** (steady decline over time)

Response distribution shows differences across departments:

- Higher participation in HR
- Moderate engagement in Production and Sales
- Lower engagement in Engineering and Finance

A **one-size-fits-all engagement strategy will not be effective**. Department-specific interventions are required.

A significant decline is observed in:

- **Workplace Relationships** (sharp drop across months)
- **Happiness levels** (steady decline over time)

Implications:

- Potential weakening of team cohesion and trust
- Reduced emotional connection to the workplace
- Possible communication or leadership gaps

Insight:
This is the most critical risk area and may negatively impact collaboration, retention, and overall morale if not addressed.

5. Recommendations & Next Steps

5.1 Standardized Survey Design

To enable automation:

- Maintain **consistent core questions across cycles**
- Allow optional rotating questions separately

5.2 Integrate Data Systems (BambooHR)

- Use API integration to connect BambooHR to Power BI
- Automate data refresh cycles

Eliminates manual data handling

5.3 Implementing a Data Model for Flexibility

- Use of **long (unpivoted) data structure**
- Mapping questions to themes dynamically

5.4 Enhance Dashboard Capabilities

- Real-time KPI tracking
- Department benchmarking
- Trend analysis across survey periods

5.5 Future Implementation Plan

Phase 1: Data Standardization

Phase 2: System Integration

Phase 3: Automation

Phase 4: Advanced Analytics

6. Appendices

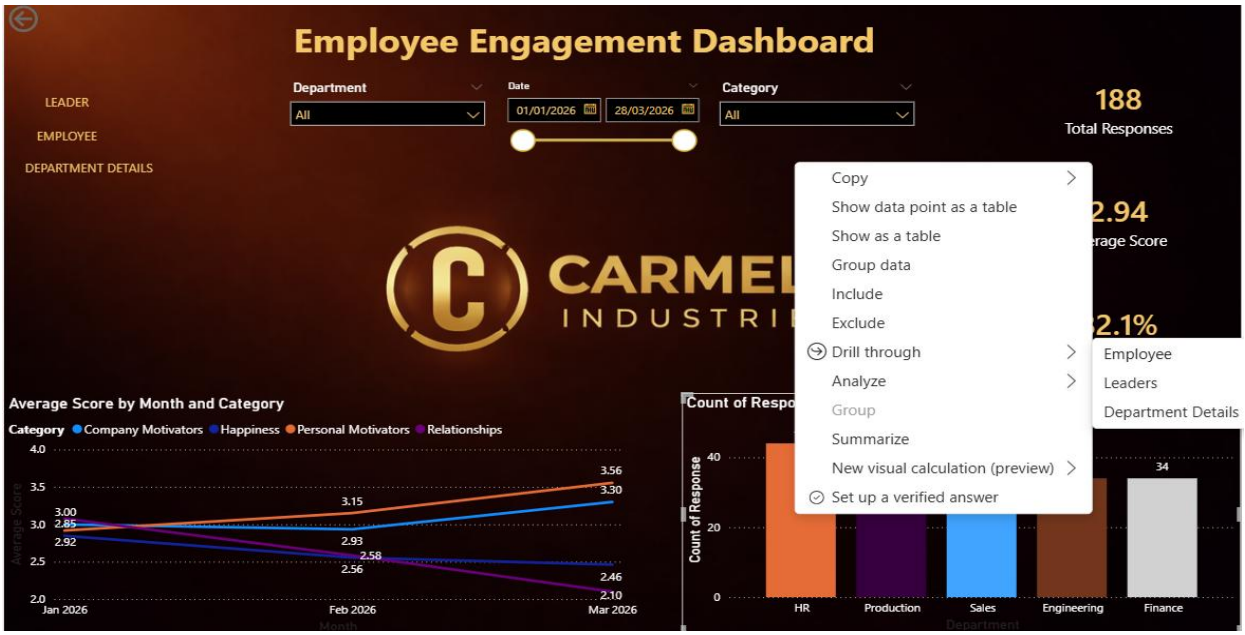
Appendix A: Data Dictionary

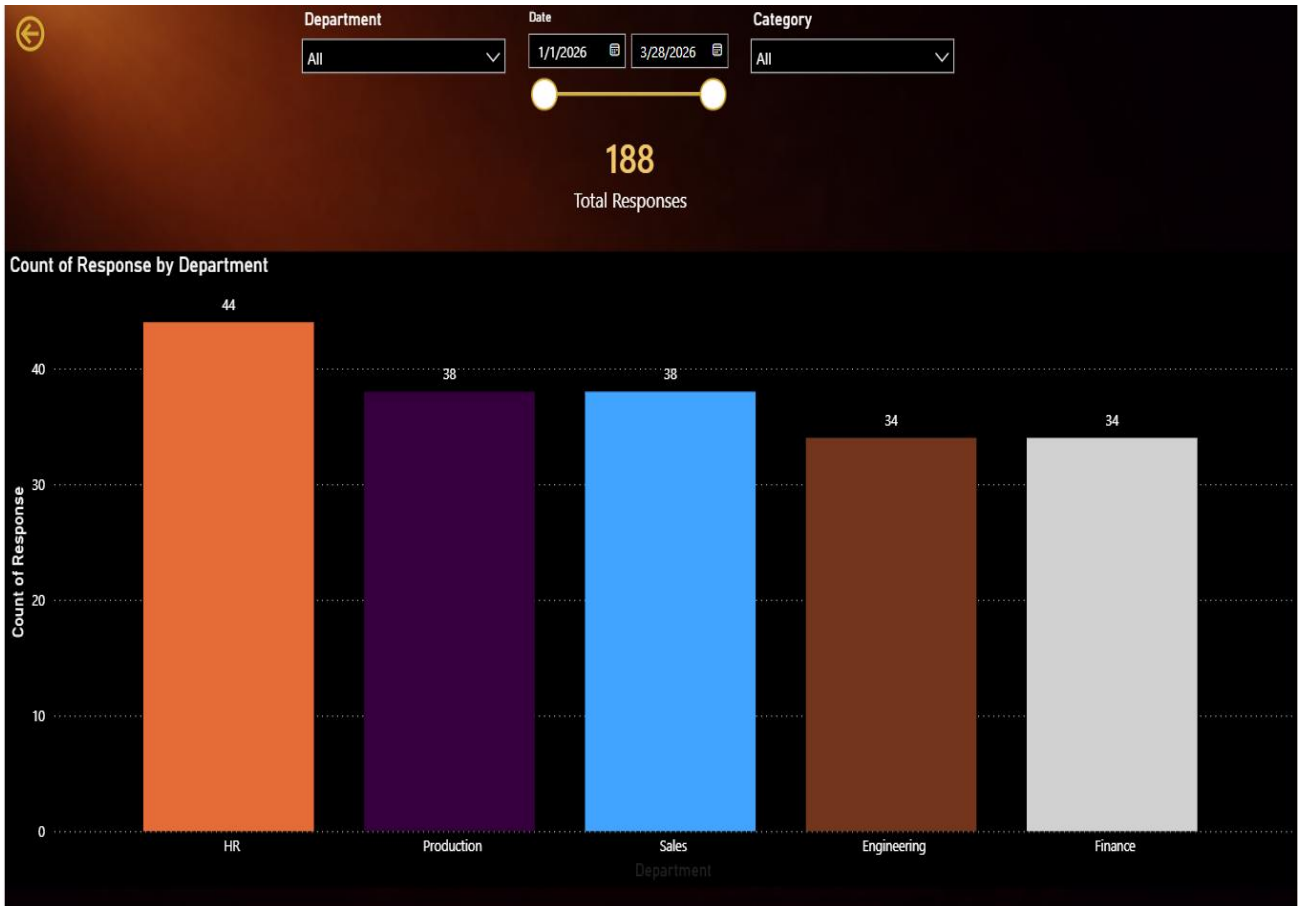
Field Name	Description	Data Type	Source
Timestamp	Date and time of survey submission	Date/Time	Google Form (Carmela industries-based questionnaire)
Department	Employee department (grouped)	Categorical	Carmela Industries
Years_in_Company	Employee tenure category	Categorical	Simulated
Happiness motivator	Derived score for happiness responses	Text	Carmela Industries theme
Personal motivator	Personal motivation level	Text	Carmela Industries theme
Company motivator	Company support perception	Text	Carmela Industries theme
Relationship motivator	Workplace relationship quality	Text	Carmela Industries theme
NPS_1	Likelihood to recommend the company	Numeric (1–5)	Simulated
Question	Selected survey response statement	Text	Carmela Industries questionnaire
Theme	Engagement dimension	Categorical	Derived
Response	Count indicator (1 per response)	Numeric	Derived

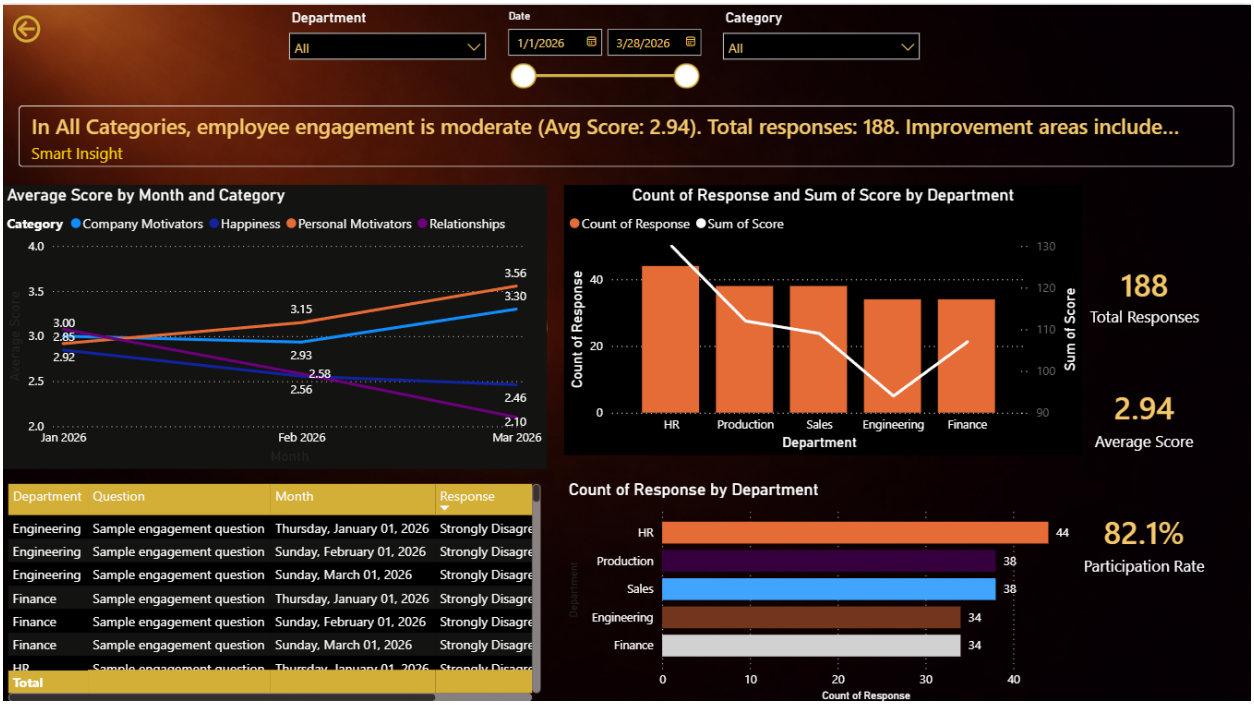
Appendix B: Data Lineage

Source → Google Forms → Excel → Power BI → Dashboard

Appendix C: Dashboard Screenshots


Employee Engagement Dashboard
LEADER
EMPLOYEE
DEPARTMENT DETAILS
Department
All
Date
01/01/2026
28/03/2026
Category
All
188
Total Responses
Copy
Show data point as a table
Show as a table
Group data
Include
Exclude
Drill through
Analyze
Group
Summarize
New visual calculation (preview)
Set up a verified answer
Employee
Leaders
Department Details
Average Score by Month and Category
Category Company Motivators Happiness Personal Motivators Relationships
Jan 2026
Feb 2026
Mar 2026
Count of Response
HR
Production
Sales
Engineering
Finance


Department
All
Date
1/1/2026
3/28/2026
Category
All
188
Total Responses
Count of Response by Department
Count of Response
HR
Production
Sales
Engineering
Finance
Department



Appendix D: Ethical Considerations

- No personal data used
- Aggregated insights only
- Transparent use of simulated dataset